Executive Summary

Context

Instacart generates revenue through two streams: transaction revenue and other revenue streams (advertising and enterprise SaaS), with the transaction revenue accounting for roughly 73% of total revenue as of December 31, 2025. Within the transaction revenue stream, gross transaction value (GTV) generated by customers placing orders was the primary driver of 2025's revenue growth over 2024. The two main ways to drive growth in GTV are either by order volume or basket size. For our analysis, we focused on two strategies to increase basket size.

We targeted basket size through two complementary surfaces: personalized sponsored ads at the top of the funnel (informed by customer segmentation) and cart-based suggestions at checkout (informed by co-purchase rules). Together, these cover the full purchase funnel, in which discovery occurs during browsing and incremental additions happen at checkout.

Approach

Following the missing-data treatment described in the notebook, we built two analyses:

Customer segmentation

We filtered to engaged users with sufficient purchase history for stable organic-share estimates, then fit GMM across various k values on organic-share features. Selected k=3 by combining BIC/AIC with bootstrap stability (50 runs, ARI), choosing interpretability over the marginal stability gain at k=4.

Association rules

We mined product co-occurrence via FP-growth on transactions pre-filtered by support (>=0.5%, multi-item baskets only), ranked rules by lift and confidence.

Findings

The three clusters based on organic preference, organic-leaning (54%), selective (28%), and conventional-leaning (18%), differ primarily in aisle preferences rather than shopping cadence or volume. These segments are differentiable enough to support distinct ad treatments, where organic-prioritized ads for the organic-leaning cluster, and conventional alternatives for the conventional-leaning cluster. Per-cluster aisle-lift identifies over-indexed categories beyond core staples, providing supplemental signal for ad decisions when ad slots extend beyond the core grocery basket.

The association rules revealed strong recipe-driven and pantry restocking patterns. The alignment with Instacart's recipe-to-cart feature suggests the association rules are surfacing real signal instead of spurious co-occurrence, supporting their use as a starting point for cart suggestion ranking. Further incrementality testing is needed to distinguish correlational lift from causal lift.

Methodological Caveats

Four caveats inform how these findings should be interpreted. First, segment proportions biases toward engaged Instacart users, not the broader grocery-shopping population, and likely overstate the organic-leaning share. Second, BIC-optimal k was inflated by GMM's Gaussian misfit on bounded organic-share features, which is why model selection combined BIC with bootstrap stability rather than relying on either alone. Third, given that customer segmentation is performed on an engaged-customer cohort, users with no or limited purchase history cannot be assigned to a segment, leaving a cold-start gap that is partially addressed by the association rules approach, which operates on the basket-in-progress and does not require historical preference data. Fourth, association rules should be treated as a ranking starting point, not a deployment-ready ranker. Without incrementality testing, top rules by lift may consume suggestion slots that would have generated equivalent purchases without the suggestion.

Recommended Next Steps

- Acquire demographic and price data to distinguish whether organic preference reflects health-consciousness, income, or other lifestyle factors, and to weight rules by transaction value.
- Mine association rules at brand level once brand metadata is available, enabling brand-affinity targeting for sponsored ads.
- Run incrementality testing on top association rules to prioritize causal recommendations over predictive ones.